INTRODUCING A SINHALA TWO-WAY COMMUNICATION SYSTEM FOR CHILDREN WITH SPEAKING DISABILITIES

25-26J-223

Final Report (Draft)

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B.Sc. (Hons) Degree in Information Technology Specialized in

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Department of Information Technology

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
HCI	Human-Computer Interaction
SSL	Sinhala Sign Language
AAC	Augmentative and Alternative Communication
UCD	User-Centered Design
UI	User Interface
UX	User Experience
RF	Random Forest
MLP	Multilayer Perceptron
XGBoost	Extreme Gradient Boosting
API	Application Programming Interface
CSV	Comma-Separated Values
NLP	Natural Language Processing
CNN	Convolutional Neural Network
IoT	Internet of Things

Table 1: Abbreviations

Declaration

I declare that this is my own work and this dissertation1 does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Signature: Date:

The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

Signature of the supervisor: Date

Abstract

Children with speech disabilities in Sri Lanka face a significant challenge in daily communication due to limited access to Sinhala Sign Language (SSL) and the lack of inclusive tools for family interaction. Although the existence of SSL, the language lacks widespread usage due to poor awareness levels, lack of training institutions, and the difficulties of teaching and learning it for children and parents respectively. Current approaches include sign-to-text/text-to-sign communication solutions which do not offer interaction support. To overcome this problem, this research study recommends and HCI based solution in the form of Sinhala two-way communication platform where the child could communicate their needs through the sign videos and images alongside with Sinhala text. Parents, on the other hand, can respond to the child in the form of simple texts or sign videos without needing any SSL knowledge. A behavior prediction model based on artificial intelligence using XGBoost was developed to evaluate communication patterns. Training of the model was carried out using 200 records comprising of 18 ordinal variables. The model achieved a mean prediction rate of 79%. Overall, the proposed solution shows a perfect combination of HCI and AI technology. While AI contributes to intelligence, the HCI contributes to the communication platform itself.

Keywords— Sinhala Sign Language, Artificial Intelligence, Sinhala Two-way Communication, Human-Computer Interaction

Acknowledgement

I would like to express my deepest gratitude to my supervisor, Ms. Thamali Dassanayake, for her continuous support, valuable insights, and guidance throughout the project. Her feedback helped me stay focused and improve the quality of my research.

I am also thankful to my co-supervisor, Ms. Thisara Shyamalee, in enhancing the research outcome.

Special thanks go to the generous cooperation of Ms. Pradeepa from Yashodara Vidyalaya in Balangoda. My sincere gratitude goes out to the children and parents of Yashodara Vidyalaya who have played a significant role in making my experiment successful. They have given me their precious time and feedback which has been very important in validating my system.

Special thanks go to the faculty of the Department of Information Technology at SLIIT for providing the resources and knowledge base that made this project possible. I appreciate the encouragement and mentorship I received throughout my academic journey.

Finally, I would like to thank my family and friends for their moral support, patience, and encouragement, which were invaluable throughout this endeavor.

1. Introduction

1.1. Background

The ability to communicate is an essential human function that facilitates social interactions, the expression of feelings, and cognition. Yet, children with speaking disorders are unable to articulate their requirements, thoughts, and emotions, causing various social, educational, and developmental problems.

In terms of the Sri Lankan scenario, the problem gets worse due to the insufficient availability of Sinhala Sign Language (SSL). Despite having a well-established system of communication through SSL, there are some reasons why it is less utilized. The unavailability of tutors, absence of adequate learning materials, particularly in rural regions, and insufficient parental education about sign language make it difficult for parents to communicate with their children, regardless of whether they know the sign language.

Digital technologies have tried to solve these problems, but most solutions only consider the sign-to-text and vice versa process or offer platforms to learn sign languages. Such technologies do not facilitate real-time interactive communication with children who are learning to speak or cannot speak at all.

1.2. Literature Survey

A review of prior work and existing systems indicates that assistive communication technologies for children with disabilities can generally be classified into three categories: sign language learning systems, translation systems, and augmentative communication systems.

Sign language learning systems involve teaching the user how to make signs using tutorials and video examples. While they serve an educational purpose, such systems do not allow real-time communication and both parties need to know the language to communicate.

Translation systems translate text into sign language and vice-versa. Though they automate part of the process, they are generally unidirectional and lack the capacity for two-way communication between the users. Also, many of them require advanced equipment or computational processing which is not practical.

Augmentative and alternative communication (AAC) systems employ symbols, pictures, or written text for communicating. They are more practical compared to the others as they are cheaper; however, the technology is not intelligent or personalized and lacks culturally adapted content. The most important deficiency with these systems is that they cannot facilitate two-way communication between children and their parents.

In summary, current systems offer only partial solutions.

1.3. Research Gap

Even though great strides have been made in improving assistive communication technologies, much room is still left for improvement with regards to addressing the communication problems faced by such children in their day-to-day lives. While many tools have already been created to help solve this problem, they tend to solve parts of the problem without giving a complete solution to it.

The most obvious limitation of the currently available tools is the one-way communication model used by them. Tools have mainly been developed to translate text into sign language and vice versa. However, while these models work well in some situations, the lack of a two-way communication system makes them practically unusable in the real world. This is because, in real life, interaction happens in two ways, especially in the case of parent-child communication.

The other significant gap that has been observed is that of limited attention towards parent-child interactions in real time. Most applications and tools have been developed for education or for the purpose of being used independently by persons with special needs. These have not paid sufficient attention to the dynamics of a family, where communication becomes an essential part of emotional support, caretaking, and activities. Thus, there are no applications created specifically to support real-time parent-child interaction.

Additionally, another key limitation in today's applications and tools can be traced in the inadequate integration of Human Computer Interaction principles, especially as applied to children. Many of the applications lack appropriate interface designs for children in the age group of 5 to 10 years old. These applications need very simple, intuitive interfaces with high visual stimulation in order to make them appealing and functional for this particular target audience.

Other than the aforementioned issues, there also appears to be a lack of intelligent assistance through Artificial Intelligence (AI) technologies. Although some current systems use Machine Learning algorithms, they tend to focus only on recognition of specific actions like detecting gestures. Very few, if any, try to recognize the behavior of the user and gain useful insights from it that can improve communication processes

Furthermore, most systems available today lack localization capabilities and cultural adaptation. They are mostly designed for international audiences and make use of globally popular languages like English. In other words, they lack the necessary capability to accommodate Sinhala language and cultural communication habits.

Therefore, one can see that all current attempts to develop an efficient communication solution are disjointed. No known systems incorporate all of the following features at once:

- Two-way communication in real time
- User-centered design based on Human-Computer Interaction principles suitable for children

- Behavior analysis using AI technologies to offer intelligent assistance
- Sinhala language and culturally appropriate modes of interaction

The purpose of this study is to fill this gap by suggesting a system that brings together all these components in one place. The combination of HCI and machine learning can help develop a single system that can be useful for children who have speech disabilities and their parents.

1.4. Research Problem

Individuals with speech impairments face a considerable amount of difficulties while trying to express their thoughts, feelings, and requirements. In comparison with other children with normally developed language, those with speech problems cannot speak; thus, they have to use gestures and facial expressions or, if possible, signs of the language. At the same time, gestures and facial expressions can be ambiguous, whereas it is quite difficult to develop a sign language system that would allow children to convey their messages.

When it comes to the problem discussed in relation to Sri Lanka, one can say that there is another difficulty that children with speech impairments may face in communicating their parents. Namely, there is not enough awareness of the existence of Sinhala Sign Language (SSL) and the possibility to use it for communication. Since the parents may neither know nor understand the signs, it may become difficult to learn the language in question.

These differences result in some negative effects. The children might feel frustrated about their incapacity to convey their thoughts properly, leading to behavioral and emotional problems. The parents, in turn, will have difficulties recognizing the needs of the child correctly and thus providing poor quality of communication. Long-term use of such ineffective communication skills is likely to cause certain cognitive and social issues in the development of the children.

There already exist various technological means of communication aimed at helping to bridge the communication gap between the parent and the child. However, none of them is able to provide a complete solution. The existing technology can be applied to teaching certain languages, like sign language, or is focused solely on one-directional communication. These means, however, cannot be used for two-way interaction, as in reality. Communication requires a constant process of giving feedback and understanding each other's point of view.

Moreover, current communication solutions do not give much attention to usability and accessibility in terms of young kids' age group. Interfaces tend to be sophisticated and textual, and also lack implementation according to the guidelines of Human-Computer Interaction (HCI). The interface of the software targeted at children within the age range between five and ten years should be intuitive and visually appealing.

Another critical shortcoming of existing solutions is the lack of intelligence. Today's systems do not implement Artificial Intelligence (AI) in order to perform behavior analysis and customize themselves according to the patterns of the conversation and users' behavior. In order to improve efficiency of communication, such solutions should have adaptive properties.

The above problems indicate the necessity for an innovative approach. This new solution should:

- Allow expressing needs and emotions of children with the help of intuitive visual means
- Let parents communicate effortlessly regardless of having or lacking previous experience with SSL
- Support two-way communication in a natural way
- Follow guidelines of HCI in order to be user-friendly and accessible
- Implement artificial intelligence algorithms

Accordingly, the research problem may be articulated as follows:

What is the process for developing a two-way Sinhala communication system designed to assist children with speech impairments that promotes effective daily communication, allows for visual communication of their needs and feelings, ensures parent-child communication in real time, and promotes understanding?

1.5. Objectives

Objectives of this study have been set to cater for the communication problems faced by children who are unable to communicate verbally. This is aimed at developing an effective system of communication that will enable them to be able to communicate effectively.

1.5.1. Main Objective

The first and foremost aim of the proposed research is to design and develop an HCI based Sinhala two-way communication system that facilitates improved communication between disabled children and their parents.

As discussed above, the proposed system will be designed keeping in view usability and accessibility requirements. This means that the developed system will have a user interface that bridges the communication gap among family members, enabling children to convey their requirements and emotions in an easy way, and parents to interpret such expressions easily. The incorporation of principles of HCI into the design of user interface will ensure that the user interface takes into account the mental capabilities and behavior patterns of children, allowing them to interact with it in a practical way.

Furthermore, the system will be intelligent enough to facilitate the communication process by taking behavioral characteristics of users into consideration.

A number of specific objectives need to be fulfilled to realize the major objective. Each specific objective covers some crucial component of the system.

- Develop a friendly visual communication interface for kids

Another important objective is to develop a user-friendly and child-friendly visual interface that does not require any training. In fact, since children with speech impairments might have problems with reading and writing, visual cues should be dominant in the interface.

The design highlights the following aspects:

- Usage of big and recognizable icons
- Categorization of icons by color to simplify navigation
- Keeping the cognitive load low

Therefore, children will be able to use the system autonomously and will not need any help from other people.

➢ Help kids express their needs via icons, sign animation, and other visual elements

The system provides different visual techniques for conveying messages. This helps to ensure that children will find an appropriate tool to convey their thoughts and desires.

This objective encompasses the following tasks:

- Visual representation of basic needs (food, games, sleeping, etc.)
- Visual communication of emotions
- Animation of signs and responses

As a result, children can better convey their ideas and intentions.

➢ Create a user-friendly interface for parents to reply through text or symbol-based responses

While this system is created primarily for the benefit of the child, it should also consider the convenience of the parents. In most cases, the parents are not fluent in sign language, hence necessitating the creation of an easily comprehensible user interface.

This objective involves:

- Creating predefined text-based responses
- Facilitating the use of text inputs
- Employing symbols to aid in communication

This will allow parents to communicate in a simple and quick manner without having to learn sign language.

➢ Create a two-way communication process

An essential part of this research is the development of a two-way communication process. Unlike most other systems, where the communication process is one-directional, communication in reality is bi-directional.

This objective revolves around:

- Allowing the child to initiate communication
- Enabling the parent to give an immediate response
- Providing feedback visually

➢ Incorporate machine learning for communication behavior analysis

In order to further improve the functionality of the system, a machine learning approach is incorporated to help with communication pattern analysis and give insight into the child's capability when communicating.

The reason behind setting this objective includes:

- Understanding strengths and weaknesses in communication skills
- Assisting in personalizing interactions with the child
- Increasing system intelligence

The XGBoost algorithm is chosen for this purpose because it provides an effective approach towards analyzing structured data. Nonetheless, it should be known that this component serves as more of a supporting role within the project.

➢ Test the system on usability, effectiveness, and performance levels

Finally, the last aim of the research is to test the proposed system in order to understand whether it fulfills the stated objectives and how well it works in practice.

This involves:

- Testing the usability of the system
- Evaluating effectiveness in improving communication skills
- Model performance testing

2. Methodology

This study incorporates the use of a mixed method whereby elements of human computer interaction design and machine learning will be used in developing the Sinhala two way communication system. In this regard, the methodology takes into account all aspects of the development process which ensures that both the user's and technical considerations have been taken care of.

These include the design process of an interactive and intelligent communication system for the purpose of easing communication between disabled children and their parents.

The major processes of this development method will take place through the systematic implementation of several stages of design and implementation.

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2.1. Approach to the Research

The methodology entails following a number of phases which will help in carrying out the research in a systematic way.

The important stages will entail:

- Requirement analysis
- System design (based on HCI)
- Data collection and processing
- Development of machine learning models
- Integration of components
- Evaluation

2.2. Requirement Analysis

During the requirement analysis phase, the needs of the target audience need to be identified, i.e., disabled kids and their parents. This is important to ensure that the development process conforms to realistic user experience.

Functional Requirements

Some of the functionalities of the system include:

- Enabling children to articulate their needs and feelings using images.
- Offering sign animations and images to enhance communication.
- Facilitating parent-child communication, where the latter can receive and reply to any inputs from their children.
- Facilitating bi-directional communication between users.
- Incorporation of machine learning to analyze user behavior.

Non-functional Requirements

Apart from functionalities, some qualities that the system should meet include:

- Usability and accessibility for young kids
- Faster response time during real-time interaction
- An easy-to-use interface
- Privacy and security of data

Such requirements provide the foundation for the system design.

2.3. System Design Approach – HCI Methodology

The system design adheres to the principles of User Centered Design (UCD), where the requirements, abilities, and limitations of end users form the crux of the process throughout the development life cycle.

The processes involved in the design are:

- Identification of user attributes (age group, ability)
- Production of low fidelity prototype
- Production of high-fidelity prototypes through Figma
- Usability evaluation and improvement of the interface

HCI principles facilitate ease-of-use and clarity of the system for children who lack communication skills.

2.4. Dataset Collection and Description

The machine learning section in the system is built upon an ordinal parent-based behavioral dataset, consisting of different behaviors observed in a child during his/her interactions and learning process.

Dataset Details

- ✓ No. of records: 200
- ✓ No. of attributes: 18
- ✓ Type of data: Ordinal (data values between 0 to 3)

The dataset contains details regarding the following:

- Ability to communicate
- Understanding of instructions
- Attention and involvement
- Learning capability
- Independence and confidence

The above attributes form a structured basis of the child's behavior.

2.5. Feature Engineering

In order to make the model more effective, we group the dataset in various categories. It will be easier to establish relationships between various behavioral traits.

Feature Categories

- ✓ Communication: The ability to communicate one's requirements and initiate interactions
- ✓ Instruction Comprehension: The ability to understand and obey instructions
- ✓ Concentration and Involvement: Time of focusing and task accomplishment
- ✓ Learning and Remembering: The ability to comprehend and remember concepts
- ✓ Independence and Self-confidence: The degree of independence and self-confidence

```python
# — Section 3: Configuration & Label Maps ——————————————————————
FEATURES = [
    "age_group","app_usage_bucket",
    "q1_familiar_instruction_response","q2_new_instruction_response",
    "q3_response_speed_visual","q4_basic_needs_expression",
    "q5_initiates_communication","q6_multi_sign_combination",
    "q7_focus_duration","q8_task_completion","q9_reminder_dependency",
    "q10_sign_retention_next_day","q11_repetition_needed",
    "q12_real_life_application","q13_routine_understanding",
    "q14_routine_response_accuracy","q15_independence_attempt",
    "q16_confidence_in_sign_usage",
]

TARGETS = [
    "communication_level",
    "instruction_understanding_level",
    "learning_retention_level",
    "attention_engagement_level",
    "independence_confidence_level",
]

TARGET_LABELS = {
    "communication_level":             {0:"Emerging",1:"Developing",2:"Functional",3:"Strong"},
    "instruction_understanding_level": {0:"Limited",1:"Improving",2:"Consistent",3:"Advanced"},
    "learning_retention_level":        {0:"Low",1:"Moderate",2:"Good",3:"Strong"},
    "attention_engagement_level":      {0:"Very Short",1:"Short",2:"Stable",3:"Sustained"},
    "independence_confidence_level":   {0:"Dependent",1:"Assisted",2:"Semi-Indep.",3:"Independent"},
}

TARGET_SHORT = {
    "communication_level":             "Communication",
    "instruction_understanding_level": "Instruction",
    "learning_retention_level":        "Learning",
    "attention_engagement_level":      "Attention",
```

Figure 1: Features and targets

2.6. Data Processing

Preprocessing is done prior to training the model.

Some of the preprocessing tasks include:

- Conversion of ordinal values to numeric form.
- Data quality checking for missing values and inconsistencies.
- Partitioning of the data into training and testing set (80/20 ratio).

- Normalization of the data where necessary.

It should be noted that these tasks are carried out before feeding the model with data.

2.7. Machine Learning Model Development

In order to select the optimal model, several machine learning models are applied and analyzed.

Models Applied

- Random Forest
- Multilayer Perceptron (MLP)
- XGBoost (optimal model)

Each of the models is trained using the created data set.

XGBoost was finally chosen as the model based on its high level of performance and ability to work efficiently with structured data.

The strengths of XGBoost are the following:

- Accurate predictions
- High level of complexity of relationships that can be learned
- Robustness against overfitting
- Efficiency when dealing with small amounts of data

XGBoost operates by building an ensemble of decision trees in which each tree corrects the mistakes of the former ones.

2.8. Model Training and Testing Data

The data is split into training and testing data to assess the performance of the models.

- ✓ Training Data: 80% of the data
- ✓ Testing Data: 20% of the data

Models are trained on training data and tested using testing data to confirm their effectiveness.

2.9. Evaluation Metrics

Performance of the machine learning models will be evaluated based on evaluation metrics.

- Accuracy: Calculates the percentage of correct predictions
- F1-Score: Determines a compromise between precision and recall
- Confusion Matrix: Evaluates errors and distribution among classes

All of these metrics will give a complete evaluation of the model’s performance.

2.10. System Integration

The last phase of the methodology is concerned with the incorporation of the machine learning algorithm within the HCI-driven framework.

These steps are included in the integration process:

- Incorporating the trained model in the backend
- Linking the user inputs to the predictor component
- Rendering the output via the user interface

3. Results and Analysis

In this part, we present the evaluation results of the machine learning algorithms used in our approach and provide an extensive analysis of their performance. As a first step, the main goal here is to determine the best performing model among the evaluated algorithms for incorporation in the Sinhala two-way communication system.

The models have been compared based on common classification metrics, namely accuracy and F1-score, to balance between different types of errors and measure the performance of predictions made by the models.

3.1.Comparison of Models' Performance

The performances of the models (Random Forest, Multilayer Perceptron (MLP), and XGBoost) are listed in the following table:

Model	Accuracy	F1 Score
Random Forest	76%	73%
MLP	77.5%	76%
XGBoost	79%	77.5%

Table 2: Comparison of model's performance

As seen from the results above, the XGBoost outperformed all other algorithms in terms of accuracy and F1-score.

3.2.Performance Interpretation

From the outcomes of the analysis, some critical findings about the performance of the models have emerged.

1. Performance of XGBoost Model

Among all other models tested in this study, XGBoost shows the best performance. The higher value of accuracy implies that the predictions of the model are accurate, whereas the high F1 score implies that the model performs well in terms of precision and recall.

The effectiveness of XGBoost can be ascribed to the fact that:

- It is capable of capturing complex associations between features
- It can improve the predictive accuracy of the models by boosting
- It has an excellent capacity for handling structured and ordinal data

Moreover, the consistency of XGBoost performance has been found in different behavioral domains such as:

- Communication ability
- Attention and engagement
- Independence and confidence

2. Performance of MLP (Neural Network)

Moderate level of performance can be observed from the MLP (Neural Network), as its results are slightly inferior to XGBoost but better than Random Forest. The mentioned performance shows that the model can effectively balance between precision and recall.

It should be noted that MLP performs well when it comes to the learning features. The reason lies in the model's ability to detect non-linear relations in the data. Nevertheless, its overall performance is limited by the following factors:

- Susceptibility to small sample sizes
- Need for parameters' adjustment
- Possibility of over- or underfittin

Despite all drawbacks mentioned above, the proposed MLP model can provide additional insights into the problem under consideration.

3. Performance of Random Forest

Moderately low level of the performance is achieved by Random Forest. It is worth mentioning that the proposed algorithm produces reasonable results and shows its advantages.

The following qualities can be considered as key strengths of Random Forest:

- Resistance to noises
- Ability to work with multiple features
- Interpretability

However, the algorithm shows lower performance rates in comparison with XGBoost due to inability to deal with feature interactions properly.

3.3. Domain-based Analysis

Besides performance metrics, the models were also analyzed based on various domains for their efficacy in each of those fields.

Analysis reveals the following:

- Communication domain: Most suitable for XGBoost owing to good modeling of interactions among features
- Instruction comprehension: Relative success of Random Forest because of logical reasoning in decisions
- Learning and retention: Higher efficiency of MLP owing to its capacity to model non-linear relationships
- Attention and engagement: Consistent predictions made by XGBoost
- Independence and confidence: Superior performance of XGBoost

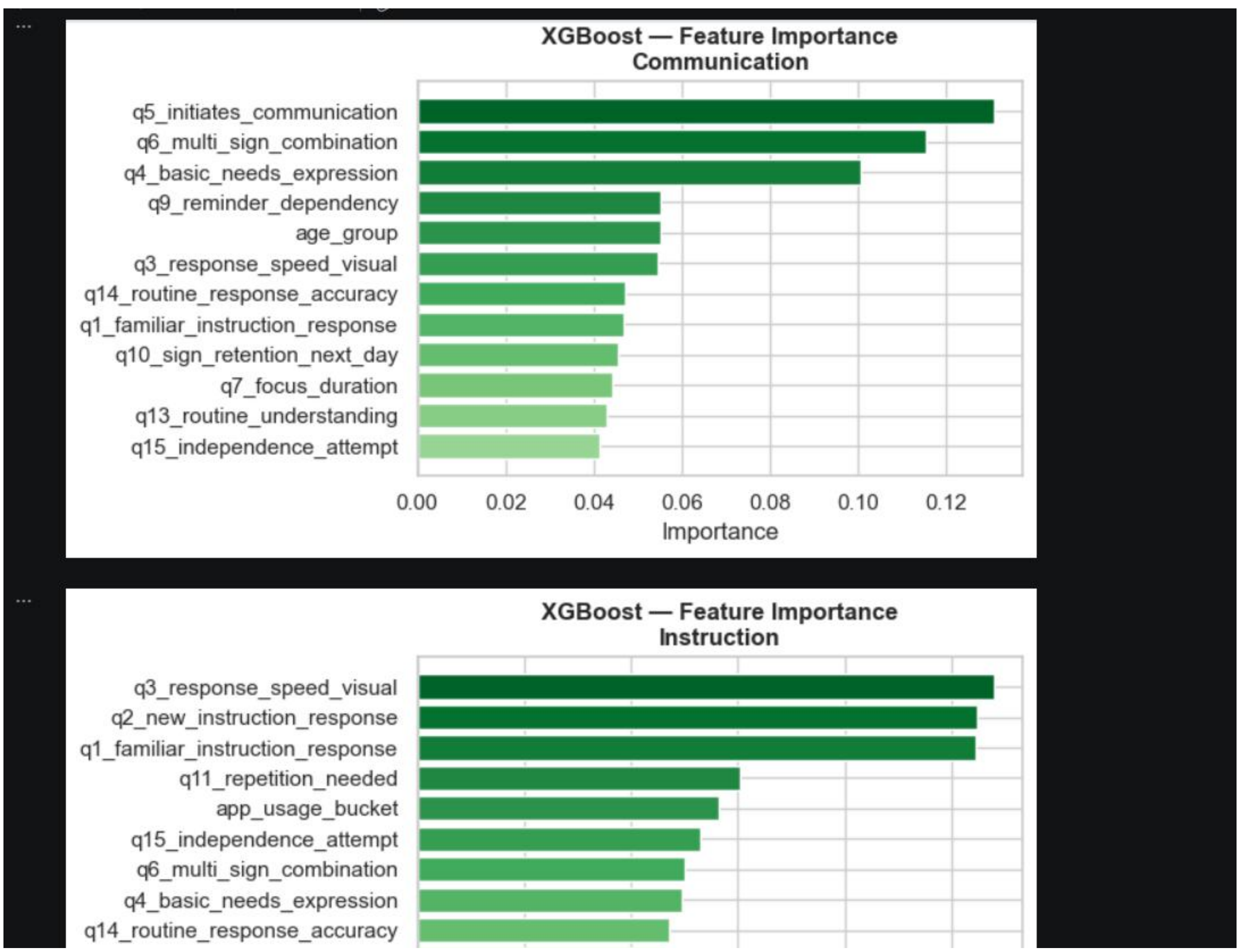


Figure 2: Feature Importance - Instruction and Communication

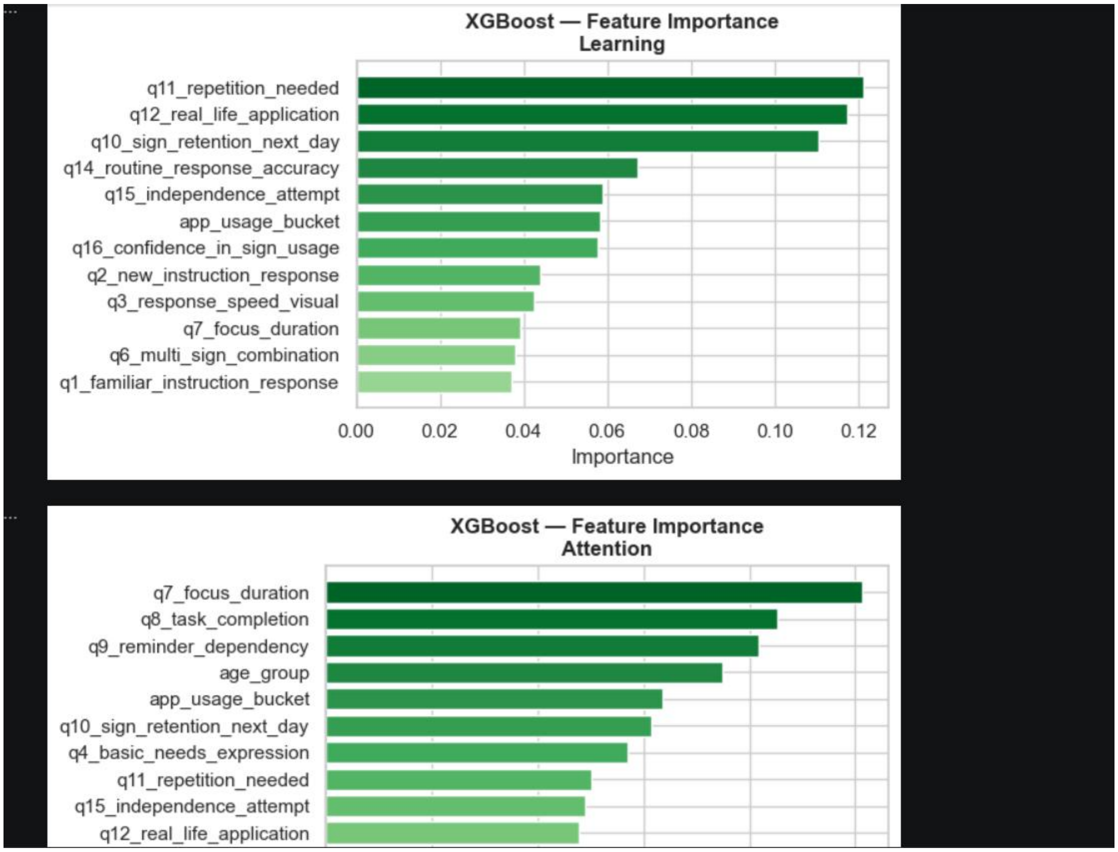
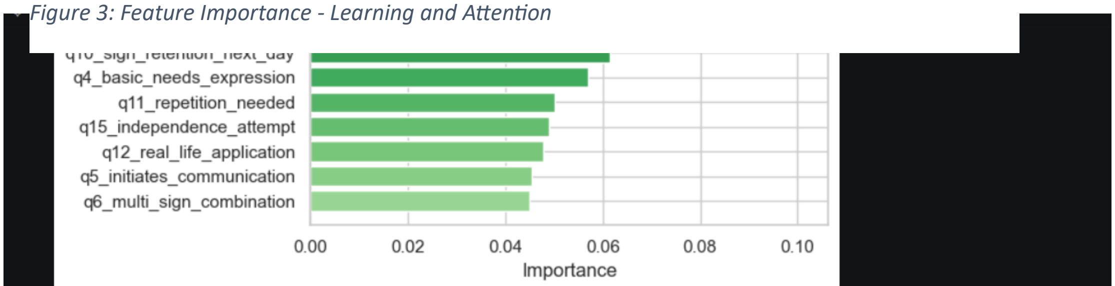


Figure 3: Feature Importance - Learning and Attention



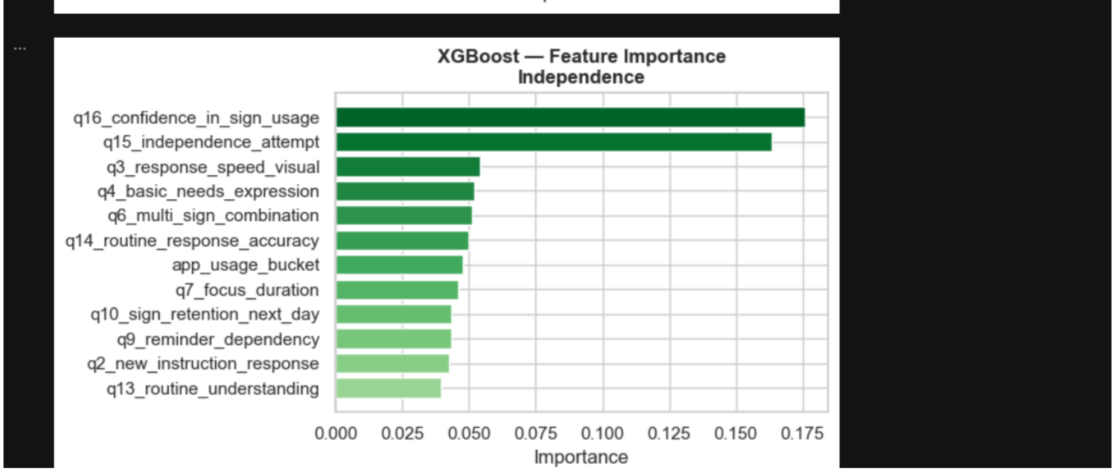


Figure 4: Feature Importance - Independence

3.4. Key Findings

Conclusion from the Analysis:

The following are some important findings from the above analysis:

- XGBoost gives the best accuracy and balance
- All the models work relatively well, suggesting that data is of good quality
- Behavior features can be used for predicting communication capability

3.5. Limitations

However, even though the models performed excellently and were quite accurate, the following limitations should be considered.

3.5.1. Limitations of the Machine Learning Model

While the machine learning part of the proposed design appears to be very efficient, there are several limitations associated with it due to the particularities of the dataset used. First, the number of observations used to train the model is small (only 200). It makes the models' generalization ability rather limited since it would not allow capturing all possible behavioral patterns.

Second, the labels used in the dataset have been generated based on some of the input variables. Therefore, the algorithm might learn artificial correlations that do not necessarily exist in the real world. In other words, the predictions might be inaccurate because they do not capture real-world communication abilities.

Finally, it should be noted that the proposed solution was tested under laboratory conditions. It means that it has not been deployed to test the algorithm in real-world environments.

3.5.2. Limitations of the whole system

Apart from the limitations associated with the machine learning algorithm, there are some additional constraints associated with the entire system.

Despite being developed in accordance with the HCI principles of usability and accessibility, the system has not been rigorously validated by means of extensive tests carried out on a large scale by a sufficiently broad group of users. Thus, the assumptions regarding usability that were used as basis for designing the system may not entirely cover the requirements of various user groups depending on their abilities.

Moreover, the existing solution allows users to convey some basic needs and intentions using predefined signs, icons, and visual cues as well as sign animations that can be used for conveying simple phrases. However, there is a limit to the complexity of communication that the existing system can handle due to the fact that it uses only such visual objects.

The system does not have any functionality related to recognizing gestures made by users in sign language or using computer vision for facilitating communication. In addition, it cannot be considered as a cloud- or mobile-based service that would allow enhancing the usability and accessibility of the system.

3.5.3. Limitations of the User Testing and Evaluation

Assessment of the system relies largely on the parameters of the performance models and considerations relating to the usability of the concept. User testing through comprehensive testing with disabled children and their parents has not been done extensively for the system.

This means that:

- Usability testing on the part of the users may not be done
- Interaction problems that occur in the real world may not have been discovered
- Satisfaction and engagement levels of users have not been gauged

Moreover, there may be ethical difficulties in testing with children, particularly those with communication challenges.

4. Discussion

In this case, the proposed Sinhala Two-Way Communication System serves as a good example of a solution to communication problems for kids with speaking disabilities. In particular, through the combination of HCI principles and machine learning, the proposed system can provide a good approach towards overcoming such issues, which are important for children and parents. This part will evaluate the strengths and weaknesses of system design, two-way communication, and a role of the machine learning tool in the general solution to the problem.

In the first place, a considerable advantage of the proposed system can be seen in its HCI-based design, which is aimed at ensuring the usability of the program. In particular, such a characteristic as accessibility is extremely high due to the design of the system for use by children between the ages of 5 and 10. In terms of the HCI concept, the interface contains visual elements and very few words. Moreover, colors and icons used in the interface enable children to interact without having any reading and writing skills.

Another valuable contribution of this project is the introduction of two-way communication. While many other applications operate using one-way interaction, the presented system provides an opportunity for constant and dynamic communication between the child and its parent. It ensures that children can communicate their desires in real-time and get quick responses from the other side, improving the quality of interaction in the long run. Furthermore, exchanging information instantly helps improve the emotional connection between children and their parents, thus enhancing the quality of communication. This makes the implemented solution especially useful in cases where parents are not around.

Finally, another unique feature of the system is the use of machine learning models, including XGBoost, for analysis and prediction of users' behavior. As proven in experiments, the developed solution is capable of delivering accurate predictions regarding communication-related parameters of the system's users, allowing us to conclude that the system has good potential in terms of intelligence. At the same time, we should pay special attention to the fact that the usage of a predictive model does not make the machine learning feature our main contribution to this field.

The proposed solution has several benefits when compared to the current assistive communication devices available. These include real-time two-way communication, improved usability due to adherence to HCI standards, and intelligent support using machine learning. However, most current solutions involve only one-way communication, have poor design, and do not involve any form of behavioral analysis. The importance of incorporating various solutions to build a more efficient system becomes evident at this point.

Some limitations associated with this project can be mentioned as well. First, the sample used in machine learning is quite small and affects the generalization of the findings. Second, the use of predefined visual stimuli limits the expression of ideas and emotions. Furthermore, there is no data regarding the application of the method in the real world, meaning there will be some difficulty when evaluating the performance of the solution.

5. Conclusion

The problem of communication faced by disabled children was effectively tackled in this research project by designing a two-way Sinhala communication aid focusing on accessibility, usability, and practicality. It was recognized by researchers that the main problem with current assistive communication devices is the inability to foster interactive communication between children and parents. Hence, this aspect can be considered one of the most prominent strengths in the proposed solution.

Another strength in the developed solution is that it applies the concepts of HCI. Ease of use, visual presentation of information, and simplification of communication have been emphasized in the system to enable children to convey their needs and feelings with the help of visual aids like icons and signs. Meanwhile, parents are given an opportunity to respond using easy and simplified techniques without the necessity of knowing Sinhala Sign Language.

The incorporation of the machine learning algorithm, particularly the XGBoost model, significantly improves the system by offering behavior insights derived from user interactions. The machine learning model exhibited a remarkable ability to predict communication-related features, suggesting its feasibility in helping improve the system. Nevertheless, it must be stressed that the significant contribution made through this research pertains to the design and implementation of the communication method using HCI principles, whereas the machine learning element constitutes a secondary feature of the system.

In summary, the proposed system has contributed greatly to the domain of assistive communication technologies, as it offers a viable solution that promotes efficient communication and improves parent-child interaction while enhancing inclusivity among children who suffer from speech disorders. Despite several drawbacks associated with the limited dataset and insufficient user testings involved, the findings have shown great promise for future developments.

6. References

[1] Unicef, Disability-Inclusive Education Practices in Sri Lanka, 2021.

[2] T. Wijesinghe, "Connect and adapt to learn and live: Deaf education in Sri Lanka," 7 May 2020. [Online]. Available: https://www.ukfiet.org/2020/connect-and-adapt-to-learn-and-live-deaf-education-in-sri-lanka/.

[3] M.U.Kannangara, S.S. Thrimahavithana, V.N.Yasoda, "Empowering the Text Based Understandability of Students with Hearing Impairments," 2019.

[4] Jing Zhao, Isabel Neto, Ana Cristina Pires, Catarina Tomé-Pires, Hugo Nicolau, "Digital Technologies for Deaf and Hard of Hearing Children: a Systematic Review, Critical Reflections, and Future Research Directions," 2025.

7. Glossary

Artificial Intelligence (AI): It is an area of computer science where intelligent systems can be made to execute tasks that typically need human intelligence.

Machine Learning (ML): Machine Learning is a branch of artificial intelligence where systems are designed to learn from data and enhance their effectiveness without being programmed explicitly.

Human Computer Interaction (HCI): Human Computer Interaction is a field of science involving research and design in interactive systems aimed at enhancing usability and user experience between computers and humans.

Sinhala Sign Language (SSL): Sinhala Sign Language is a form of visual communication used by the deaf and speech-impaired communities in Sri Lanka, where communication involves using hand signals, facial expressions, and body movements.

Two-way Communication System: It is a form of interactive communication involving two parties exchanging information in real-time.

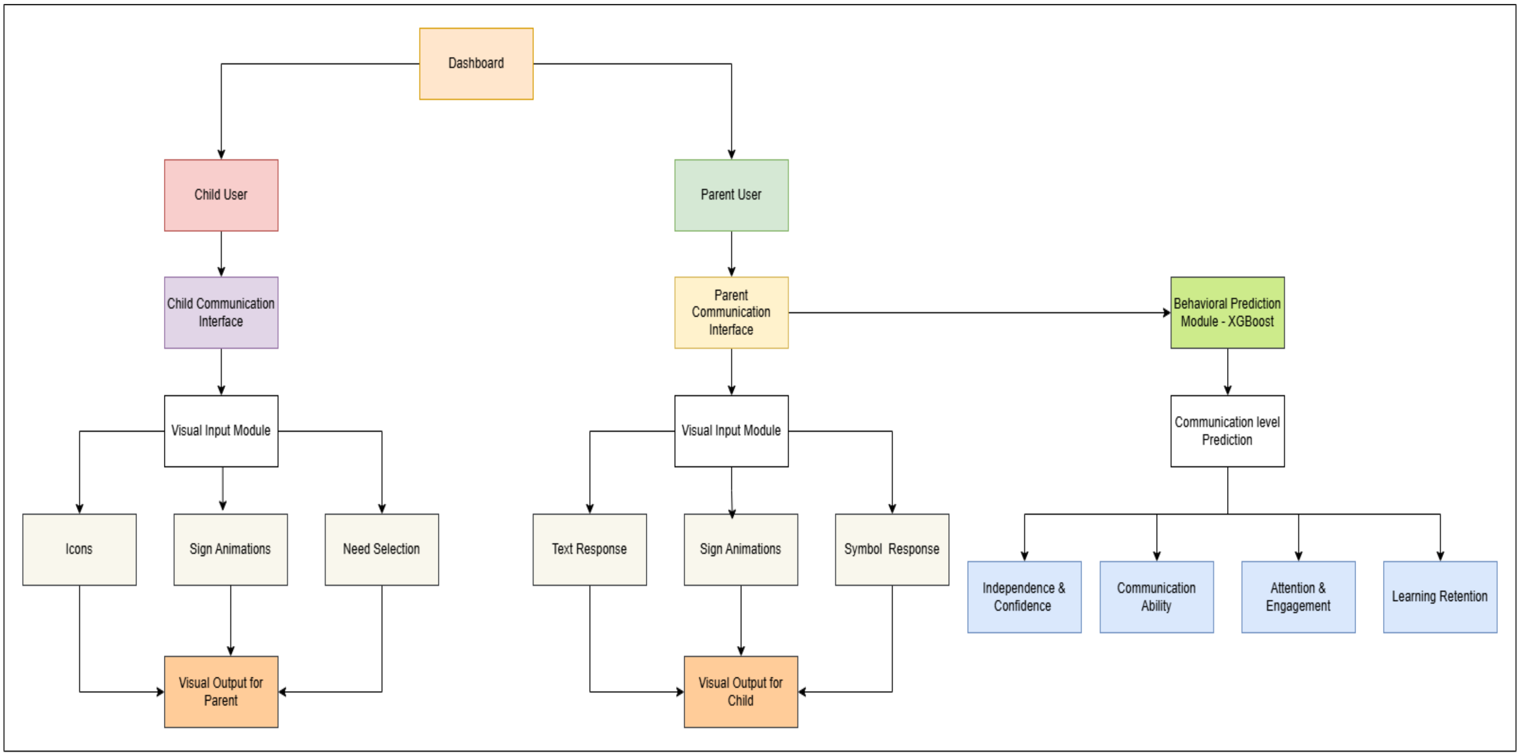
User Interface (UI): The visual layout through which users communicate with the system is known as user interface.

User Experience (UX): It is the overall user experience and satisfaction while communicating with the system.

User Centered Design (UCD): User Centered Design refers to designing systems based on user requirements and limitations.

8. Appendices

8.1. Appendix – A: System Architecture Diagram



8.2. Appendix – B: Column Mappings for targes and features

C: > Users > Asanga De Silva > Desktop > researchProject > train > {} column_mapping.json > ...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{
  "metadata": {
    "description": "Parent-reported behavioral dataset for deaf children (ages 5–10). All values are ordinal, non-diagnostic, and used for AI-driven behavioral insights.",
    "scale": "0–3 ordinal scale unless otherwise noted"
  },

  "input_columns": {
    "age_group": {
      "type": "ordinal",
      "mapping": {
        "0": "Age 5–6",
        "1": "Age 7–8",
        "2": "Age 9–10"
      }
    },

    "app_usage_bucket": {
      "type": "ordinal",
      "mapping": {
        "0": "Less than 10 minutes per day",
        "1": "10–20 minutes per day",
        "2": "More than 20 minutes per day"
      }
    },

    "q1_familiar_instruction_response": {
      "section": "Instruction Understanding",
      "mapping": {
        "0": "Does not respond",
        "1": "Responds after help",
        "2": "Responds independently",
        "3": "Responds quickly and confidently"
      }
    },

    "q2_new_instruction_response": {
      "section": "Instruction Understanding",
```

C: > Users > Asanga De Silva > Desktop > researchProject > train > {} column_mapping.json > ...

```json
  "input_columns": {
    },

    "q2_new_instruction_response": {
      "section": "Instruction Understanding",
      "mapping": {
        "0": "Is confused",
        "1": "Understands after repetition",
        "2": "Understands with visual hints",
        "3": "Understands quickly"
      }
    },

    "q3_response_speed_visual": {
      "section": "Instruction Understanding",
      "mapping": {
        "0": "Very slow",
        "1": "Slow",
        "2": "Average",
        "3": "Fast"
      }
    },

    "q4_basic_needs_expression": {
      "section": "Communication & Expression",
      "mapping": {
        "0": "Not yet",
        "1": "With help",
        "2": "Mostly independently",
        "3": "Fully independently"
      }
    },

    "q5_initiates_communication": {
      "section": "Communication & Expression",
      "mapping": {
        "0": "Never",
```

C: > Users > Asanga De Silva > Desktop > researchProject > train > {} column_mapping.json > ...

```json
  "input_columns": {
    "q4_basic_needs_expression": {
      }
    },

    "q5_initiates_communication": {
      "section": "Communication & Expression",
      "mapping": {
        "0": "Never",
        "1": "Rarely",
        "2": "Often",
        "3": "Very often"
      }
    },

    "q6_multi_sign_combination": {
      "section": "Communication & Expression",
      "mapping": {
        "0": "Never",
        "1": "Sometimes",
        "2": "Often",
        "3": "Always"
      }
    },

    "q7_focus_duration": {
      "section": "Attention & Engagement",
      "mapping": {
        "0": "Less than 1 minute",
        "1": "1–3 minutes",
        "2": "3–5 minutes",
        "3": "More than 5 minutes"
      }
    },

    "q8_task_completion": {
      "section": "Attention & Engagement",
```

Ln 1, Col 1 Spaces: 2 UTF-8 LF {} JSON

```
"input_columns": {
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    }
  },

  "q9_reminder_dependency": {
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    "note": "Higher value means less dependency (better)",
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      "1": "Often needs reminders",
      "2": "Sometimes needs reminders",
      "3": "Rarely needs reminders"
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  "q10_sign_retention_next_day": {
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      "1": "Some repetitions (3–4)",
      "2": "Few repetitions (1–2)",
      "3": "Learns quickly"
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  },

  "q12_real_life_application": {
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8.3. Appendix – C: Dataset

C: > Users > Asanga De Silva > Desktop > researchProject > train > parent_behavior_dataset.csv > data

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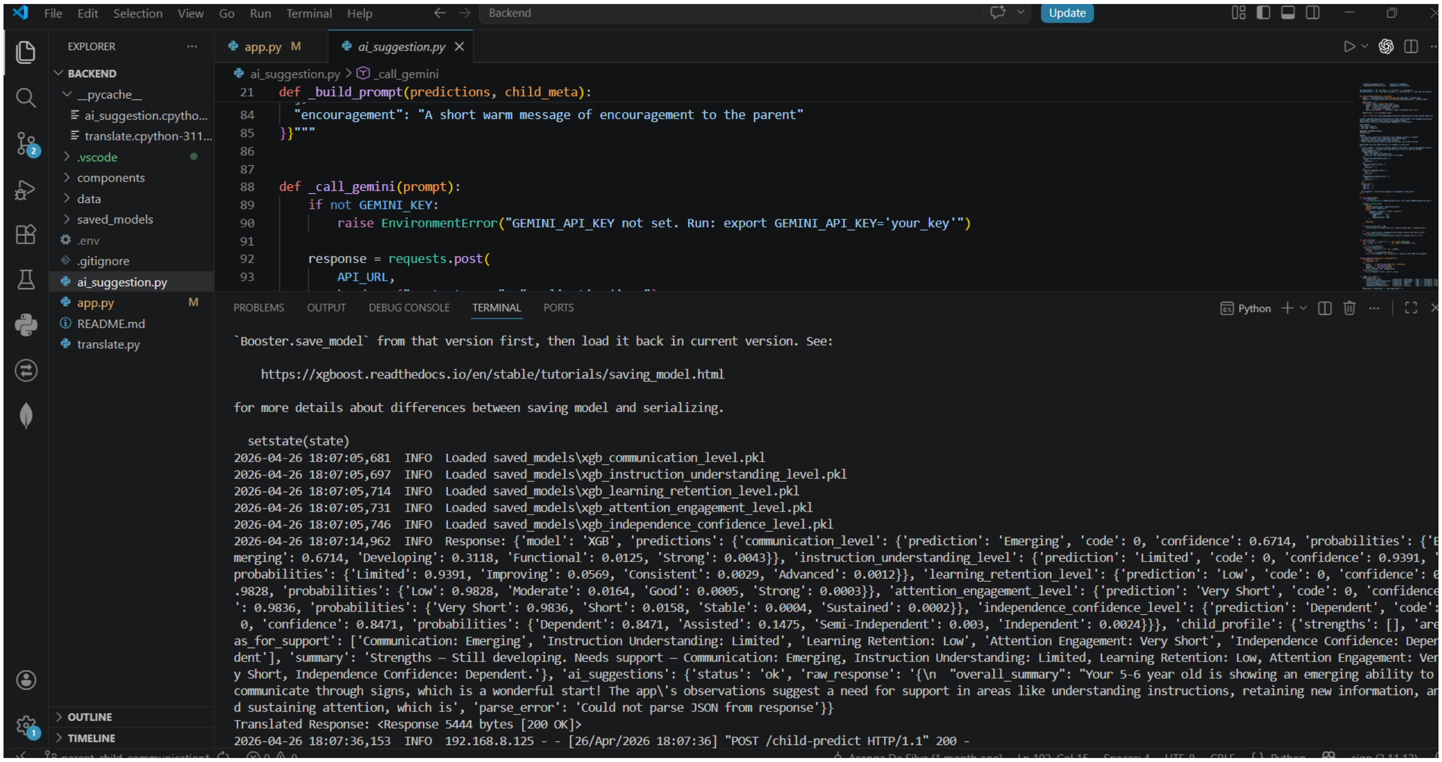
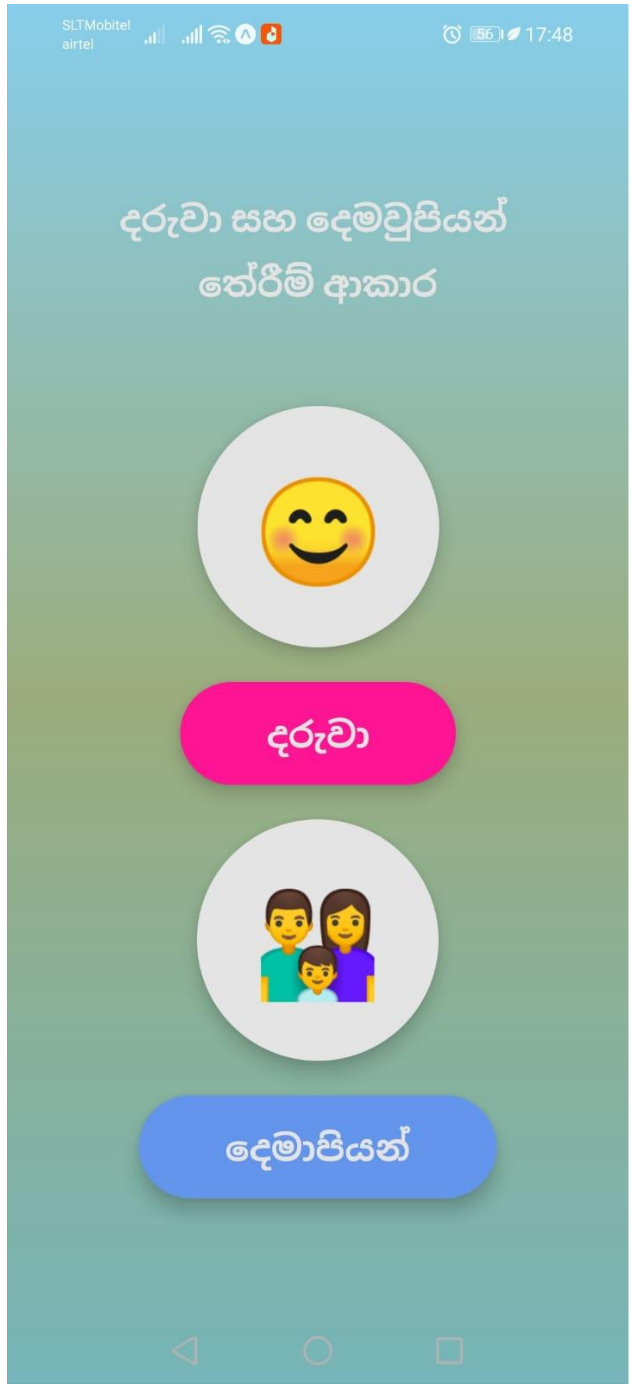
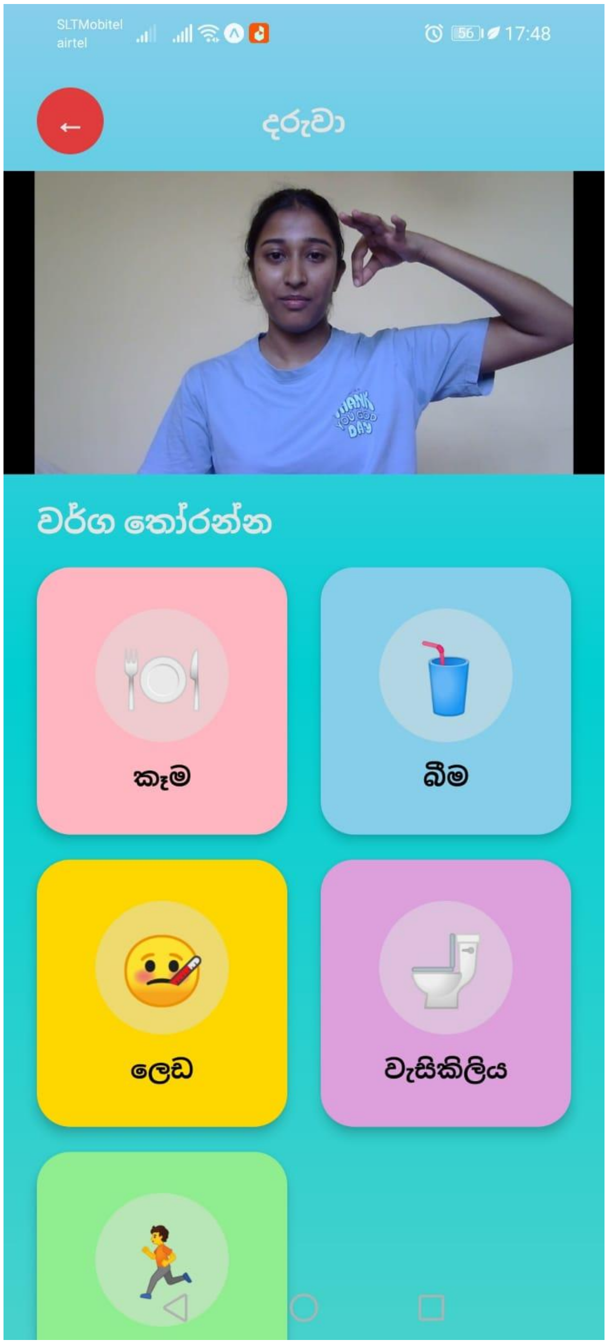
8.4. Appendix – D: Model Prediction



8.5. Appendix – E: Application – Interfaces

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වර්ග තෝරන්න
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මට පාන් කන්න ඕනේ
මට ඉදිආප්ප කන්න ඕනේ


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17:49
දෙමාපියන්
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සංකේත භාෂා වීඩියෝ බැලීමට කොටස් මත තට්ටු කරන්න
වර්ග තෝරන්න
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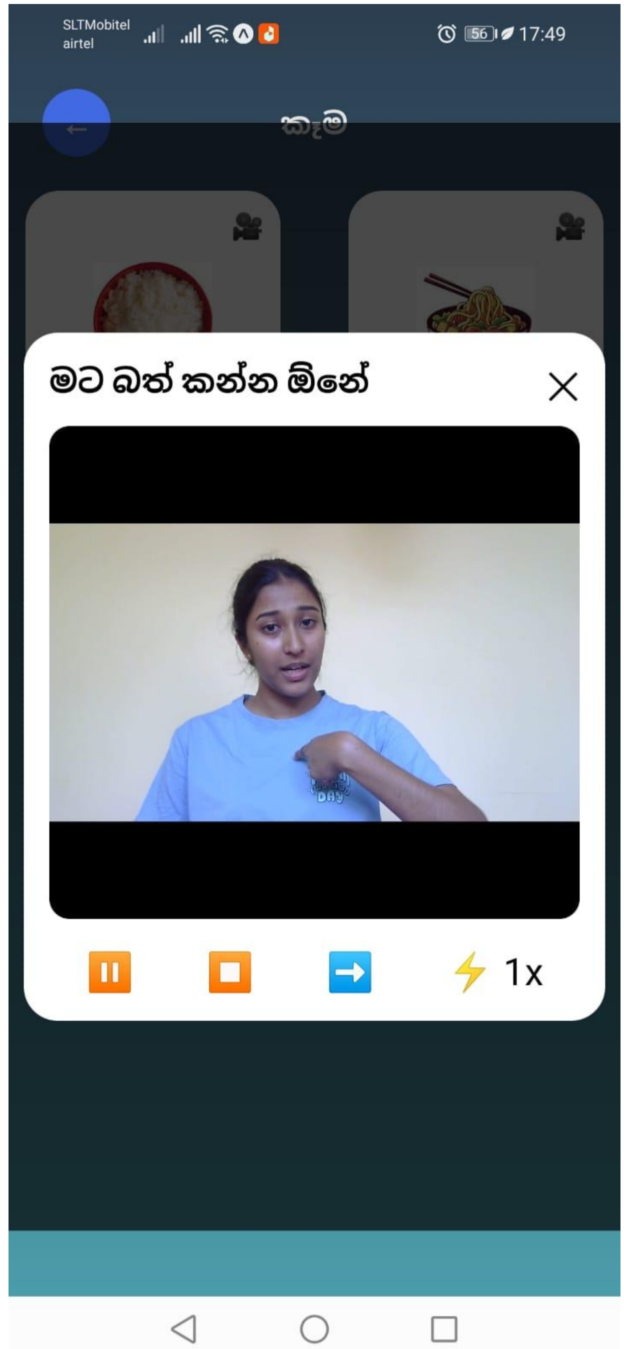
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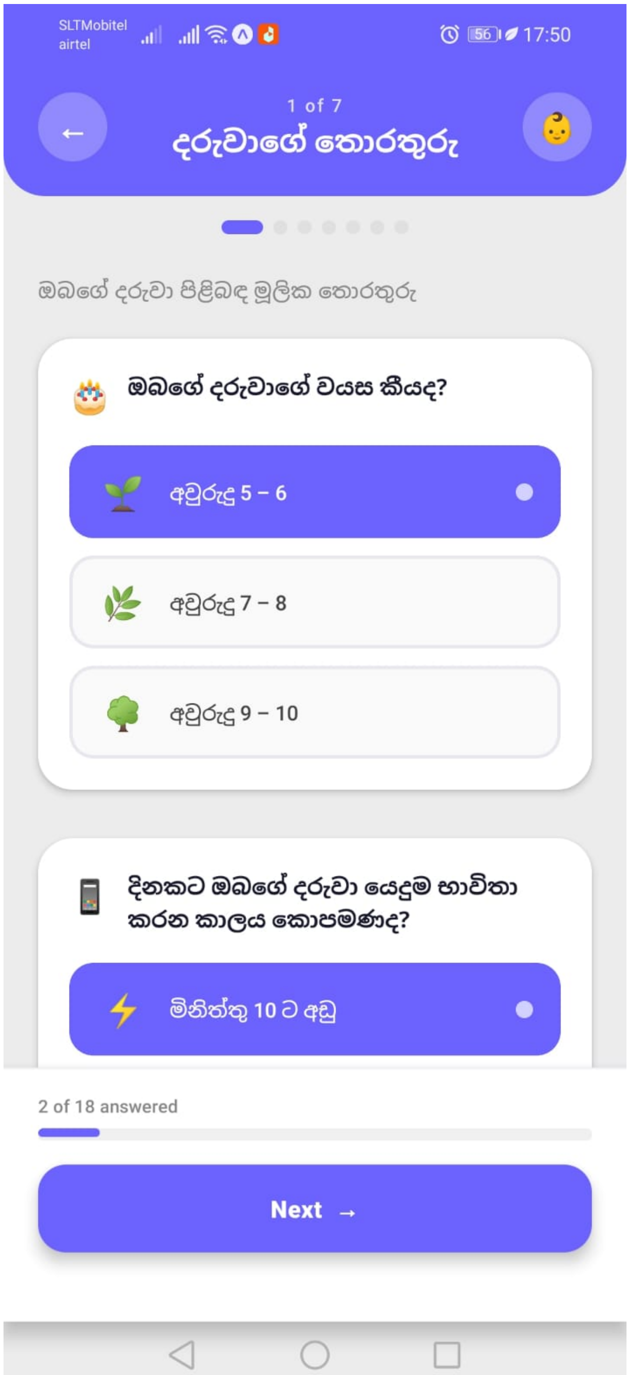
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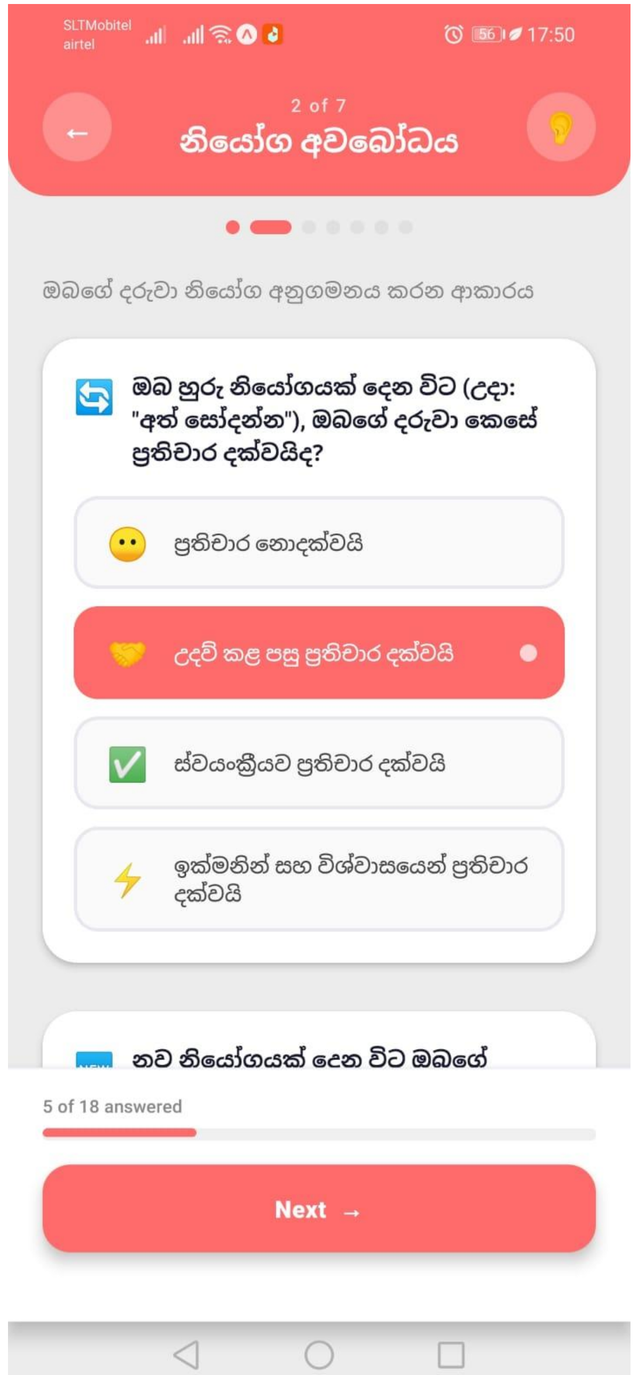
දරුවාගේ තොරතුරු
ඔබගේ දරුවා පිළිබඳ මූලික තොරතුරු
ඔබගේ දරුවාගේ වයස කීයද?
අවුරුදු 5 – 6
අවුරුදු 7 – 8
අවුරුදු 9 – 10
දිනකට ඔබගේ දරුවා යෙදුම භාවිතා කරන කාලය කොපමණද?
මිනිත්තු 10 ට අඩු
2 of 18 answered
Next →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airtel
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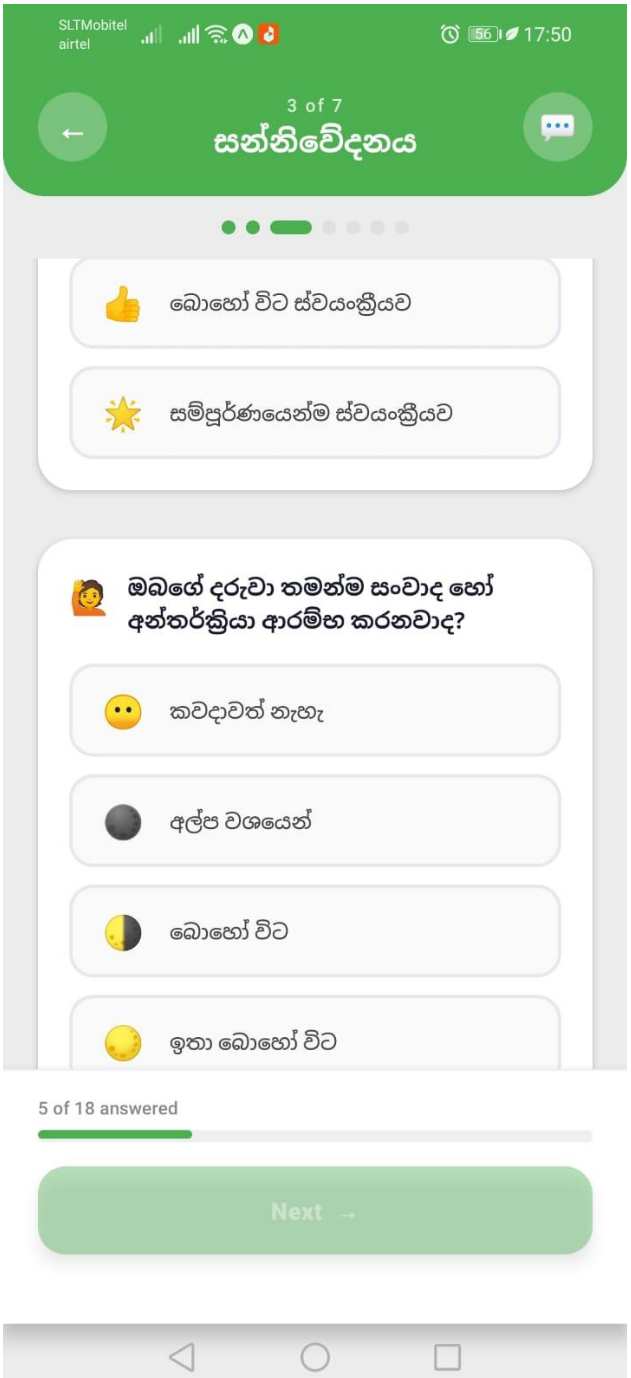
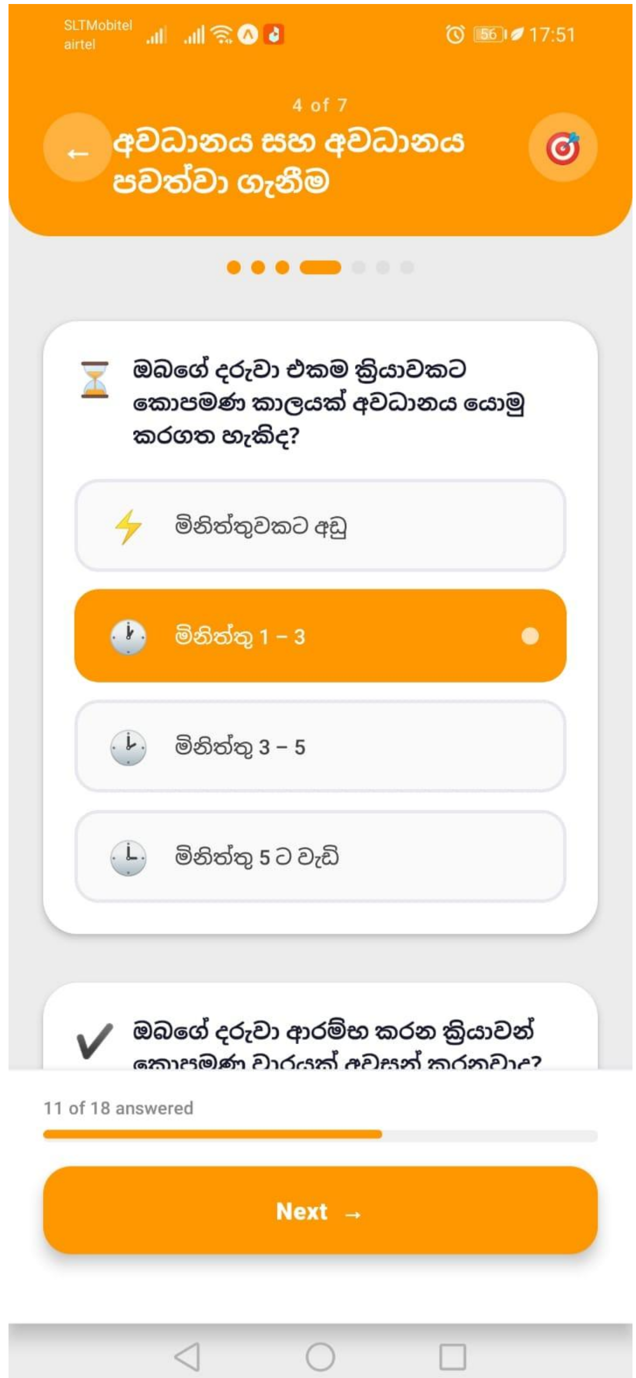
නියෝග අවබෝධය
ඔබගේ දරුවා නියෝග අනුගමනය කරන ආකාරය
ඔබ හුරු නියෝගයක් දෙන විට (උදා: "අත් සෝදන්න"), ඔබගේ දරුවා කෙසේ ප්‍රතිචාර දක්වයිද?
ප්‍රතිචාර නොදක්වයි
උදව් කළ පසු ප්‍රතිචාර දක්වයි
ස්වයංක්‍රීයව ප්‍රතිචාර දක්වයි
ඉක්මනින් සහ විශ්වාසයෙන් ප්‍රතිචාර දක්වයි
නව නියෝගයක් දෙන විට ඔබගේ
5 of 18 answered
Next →


SLTMobitel
airtel
17:50
3 of 7
සන්නිවේදනය
බොහෝ විට ස්වයංක්‍රීයව
සම්පූර්ණයෙන්ම ස්වයංක්‍රීයව
ඔබගේ දරුවා තමන්ම සංවාද හෝ අන්තර්ක්‍රියා ආරම්භ කරනවාද?
කවදාවත් නැහැ
අල්ප වශයෙන්
බොහෝ විට
ඉතා බොහෝ විට
5 of 18 answered
Next →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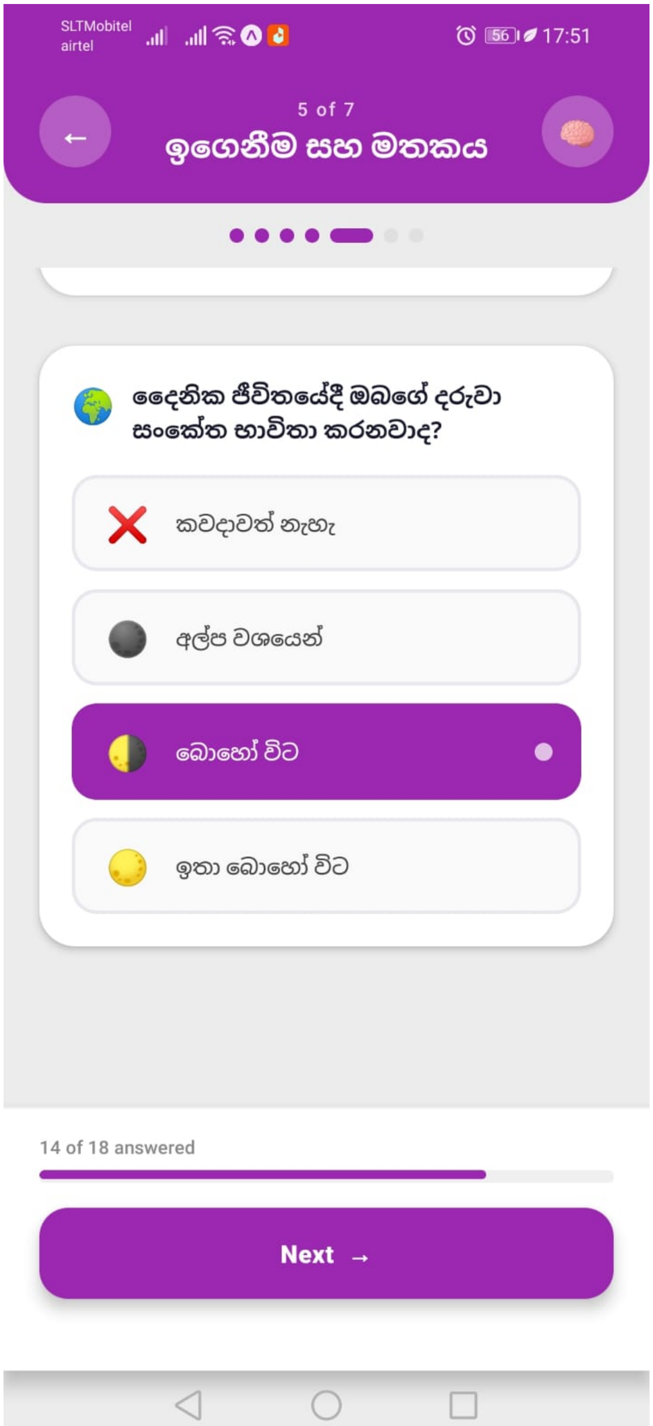
4 of 7
අවධානය සහ අවධානය පවත්වා ගැනීම
ඔබගේ දරුවා එකම ක්‍රියාවකට කොපමණ කාලයක් අවධානය යොමු කරගත හැකිද?
මිනිත්තුවකට අඩු
මිනිත්තු 1 – 3
මිනිත්තු 3 – 5
මිනිත්තු 5 ට වැඩි
ඔබගේ දරුවා ආරම්භ කරන ක්‍රියාවන්
11 of 18 answered
Next →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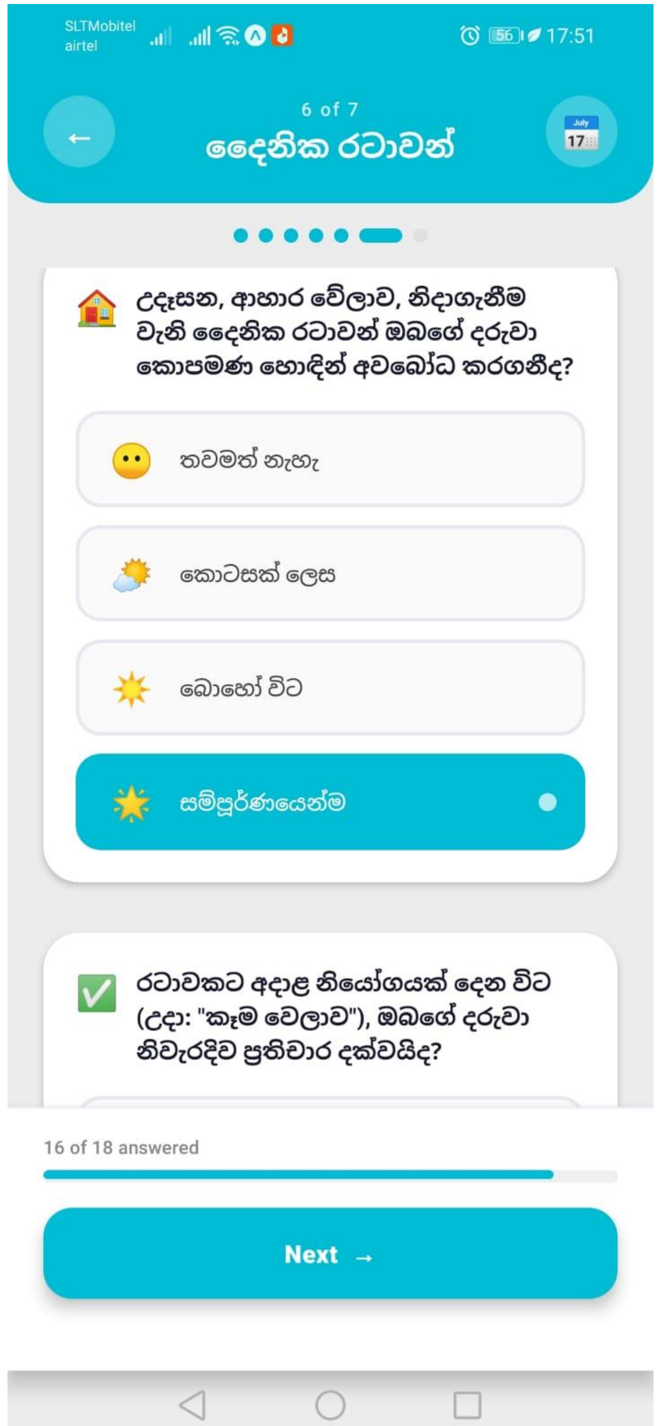
5 of 7
ඉගෙනීම සහ මතකය
දෛනික ජීවිතයේදී ඔබගේ දරුවා සංකේත භාවිතා කරනවාද?
කවදාවත් නැහැ
අල්ප වශයෙන්
බොහෝ විට
ඉතා බොහෝ විට
14 of 18 answered
Next →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17:51
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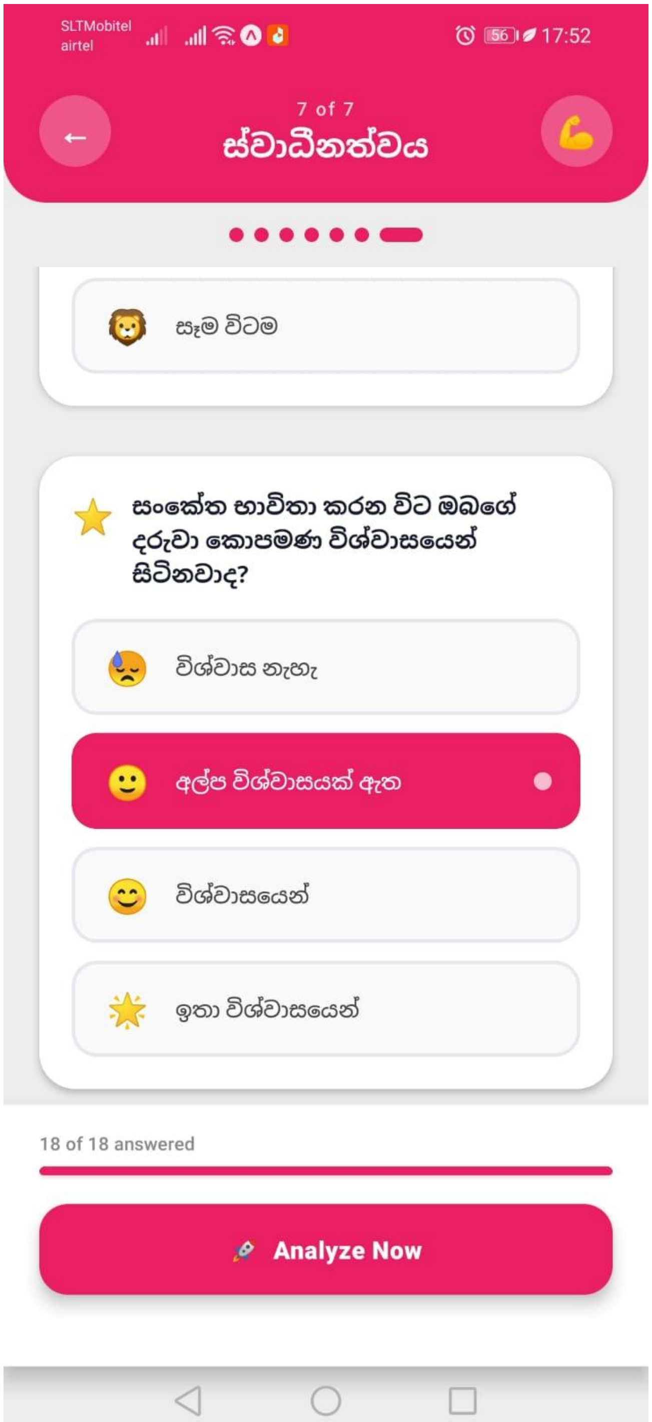
දෛනික රටාවන්
උදෑසන, ආහාර වේලාව, නිදාගැනීම වැනි දෛනික රටාවන් ඔබගේ දරුවා කොපමණ හොඳින් අවබෝධ කරගනීද?
තවමත් නැහැ
කොටසක් ලෙස
බොහෝ විට
සම්පූර්ණයෙන්ම
රටාවකට අදාළ නියෝගයක් දෙන විට (උදා: "කෑම වෙලාව"), ඔබගේ දරුවා නිවැරදිව ප්‍රතිචාර දක්වයිද?
16 of 18 answered
Next →



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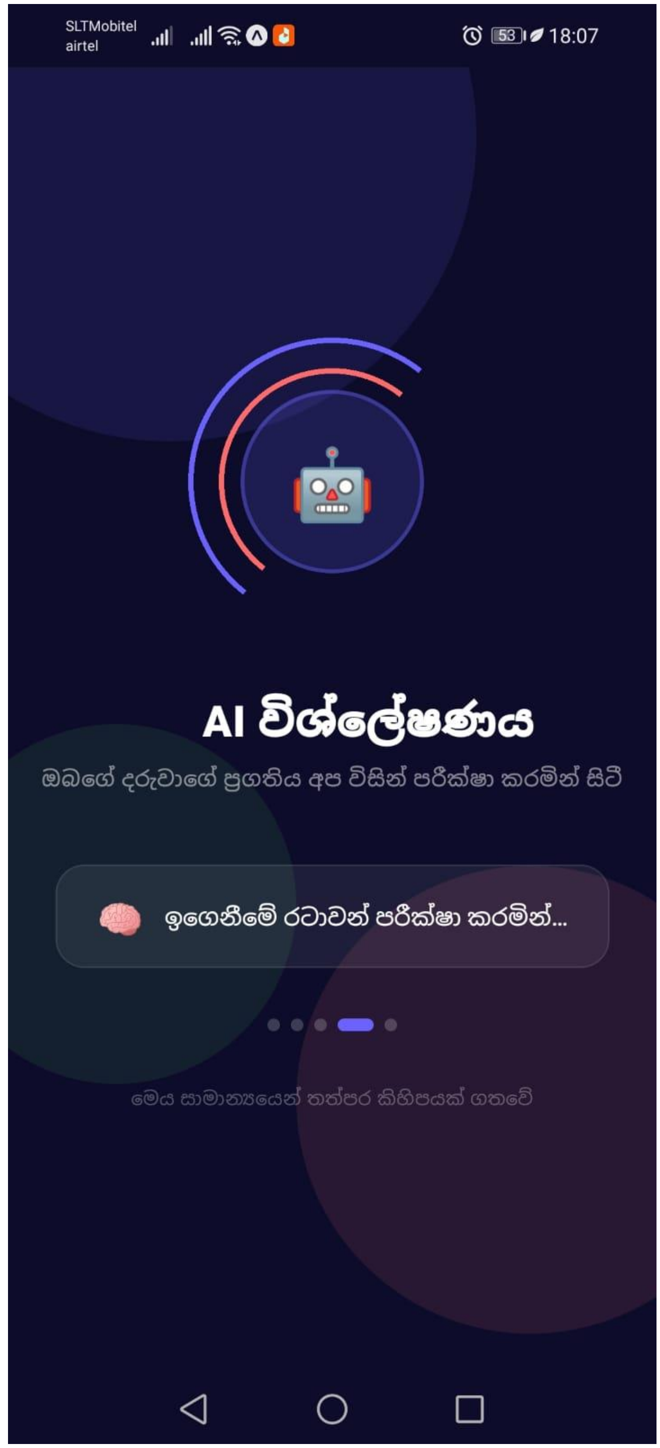
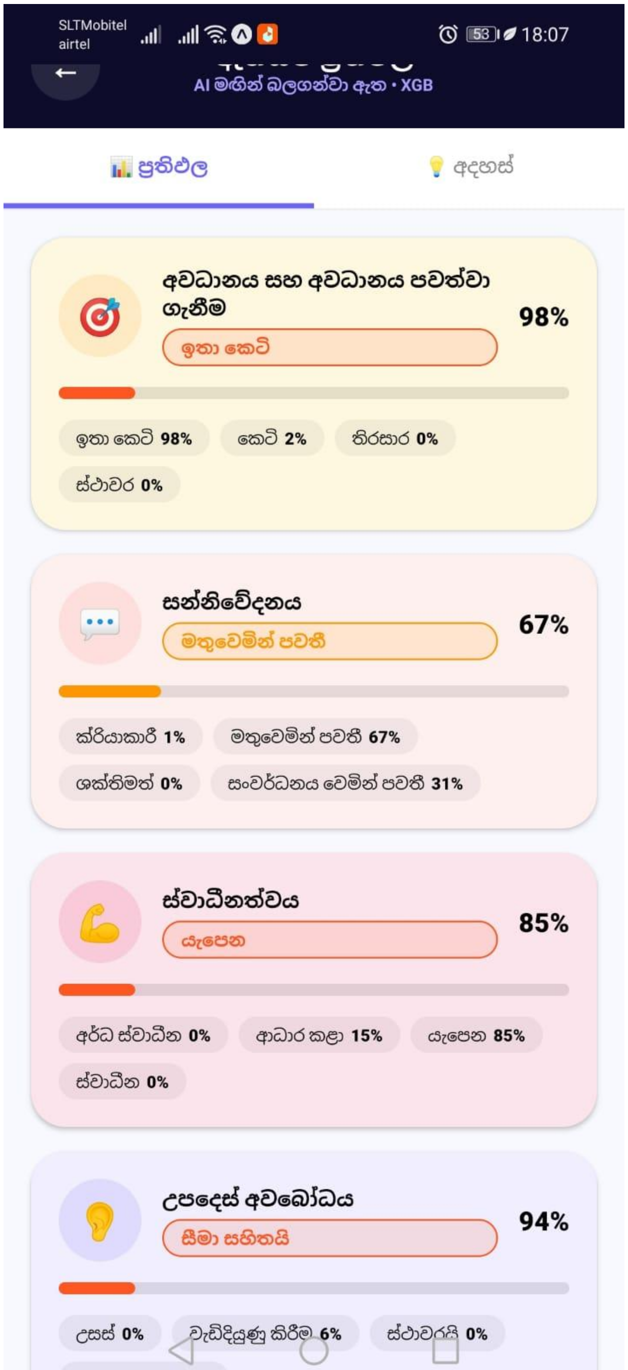
ස්වාධීනත්වය
සෑම විටම
සංකේත භාවිතා කරන විට ඔබගේ දරුවා කොපමණ විශ්වාසයෙන් සිටිනවාද?
විශ්වාස නැහැ
අල්ප විශ්වාසයක් ඇත
විශ්වාසයෙන්
ඉතා විශ්වාසයෙන්
18 of 18 answered
Analyze Now



AI විශ්ලේෂණය
ඔබගේ දරුවාගේ ප්‍රගතිය අප විසින් පරීක්ෂා කරමින් සිටී
ඉගෙනීමේ රටාවන් පරීක්ෂා කරමින්...
මෙය සාමාන්‍යයෙන් තත්පර කිහිපයක් ගතවේ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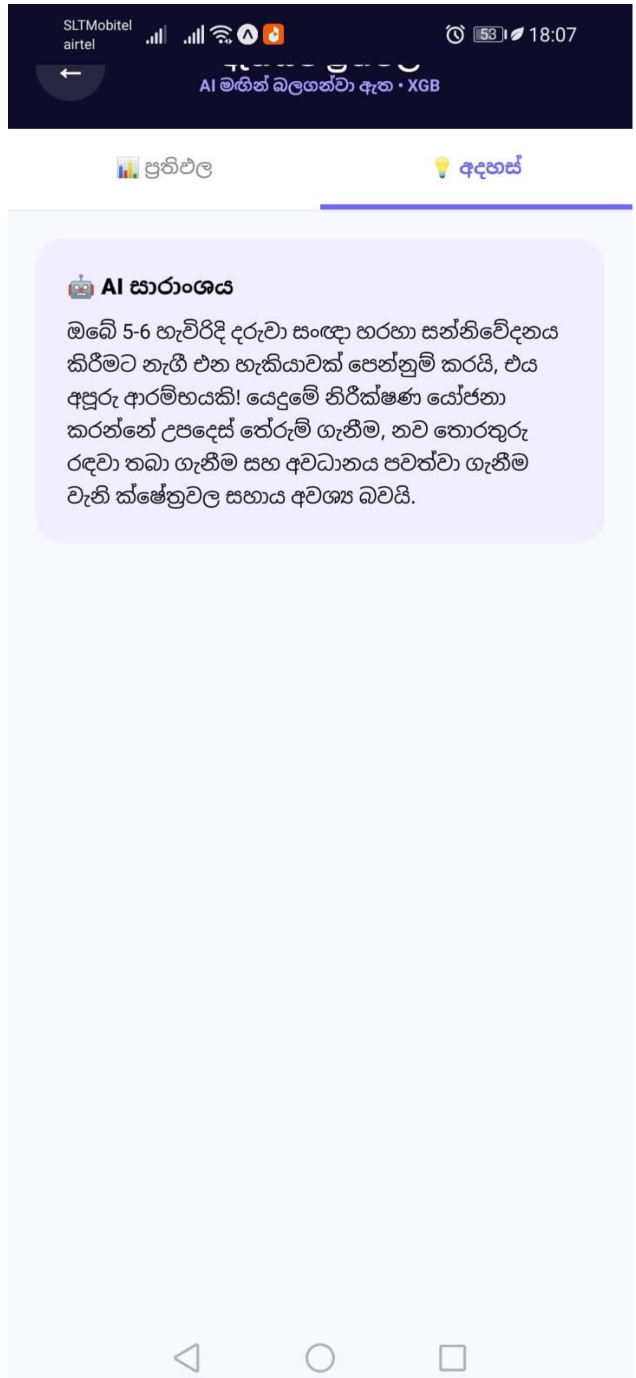
AI මඟින් බලගන්වා ඇත • XGB
📊 ප්‍රතිඵල
💡 අදහස්
අවධානය සහ අවධානය පවත්වා ගැනීම
ඉතා කෙටි
98%
ඉතා කෙටි 98%
කෙටි 2%
තිරසාර 0%
ස්ථාවර 0%
සන්නිවේදනය
මතුවෙමින් පවතී
67%
ක්‍රියාකාරී 1%
මතුවෙමින් පවතී 67%
ශක්තිමත් 0%
සංවර්ධනය වෙමින් පවතී 31%
ස්වාධීනත්වය
යැපෙන
85%
අර්ධ ස්වාධීන 0%
ආධාර කළා 15%
යැපෙන 85%
ස්වාධීන 0%
උපදෙස් අවබෝධය
සීමා සහිතයි
94%
උසස් 0%
වැඩිදියුණු කිරීම 6%
ස්ථාවරයි 0%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AI මඟින් බලගන්වා ඇත • XGB
📊 ප්‍රතිඵල
💡 අදහස්
🤖 AI සාරාංශය
ඔබේ 5-6 හැවිරිදි දරුවා සංඥා හරහා සන්නිවේදනය කිරීමට නැගී එන හැකියාවක් පෙන්නුම් කරයි, එය අපූරු ආරම්භයකි! යෙදුමේ නිරීක්ෂණ යෝජනා කරන්නේ උපදෙස් තේරුම් ගැනීම, නව තොරතුරු රඳවා තබා ගැනීම සහ අවධානය පවත්වා ගැනීම වැනි ක්ෂේත්‍රවල සහාය අවශ්‍ය බවයි.